Finite-Deformation Macroscopic Stress Relaxation for Dislocation Core Simulations

Kinematically Consistent Upper-Triangular Formalism, Spatial Tangent Push-Forward, and Nested Newton-Raphson Solvers

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October 23, 2026

Abstract

When an isolated edge dislocation is introduced into a finite crystal, standard Dirichlet (frozen mantle) boundary conditions induce spurious macroscopic reaction stresses $(\bar{\sigma}_{xx}, \bar{\sigma}_{yy}, \bar{\sigma}_{xy})$ due to non-vanishing image forces and the net insertion of an atomic half-plane. This document presents the complete mathematical formulation, kinematic justification, and computational architecture of the *Full Cauchy Tensor* macroscopic stress relaxation method implemented for 2D nonlinear lattice triangulation. To preserve the glide line geometry without artificial crystallographic rotation, we parameterize the macroscopic deformation gradient as an upper-triangular tensor $\mathbf{F}_{\text{ext}} = \begin{pmatrix} \lambda_x & \gamma \\ 0 & \lambda_y \end{pmatrix}$ with $(\mathbf{F}_{\text{ext}})_{21} \equiv 0$. The macroscopic residual Cauchy stress is iteratively relaxed to machine zero via a quadratic Newton-Raphson scheme driven by the exact fourth-order spatial tangent stiffness $\mathbb{C}^{\text{spatial}}(\mathbf{F}_{\text{ext}})$ computed via tensor contraction and push-forward of the internal acoustic tensor. We formulate the nested two-scale minimization algorithm, the core-centered coordinate update, and the subsequent topological Delaunay reconnection on the stress-free state.

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1 Introduction and Physical Problem

In computational continuum mechanics and atomistic-to-continuum multiscale modeling of crystalline defects, an edge dislocation is conventionally introduced into a finite domain $\Omega \subset \mathbb{R}^2$ by prescribing the analytical linear elastic displacement field $\mathbf{u}_0(\mathbf{X})$ (e.g., isotropic Volterra or anisotropic Stroh/Eshelby solution) across all atomic positions $\mathbf{X}_i \in \Omega$.

To isolate the nonlinear dislocation core from the outer boundary, the simulation cylinder is partitioned into two regions:

1. **Internal Core Region ($\|\mathbf{X}_i - \mathbf{X}_{\mathbf{core}}\| \leq R_{\mathbf{free}}$)**: Nodes are free to displace under energy minimization to resolve the discrete lattice core structure, nonlinear stacking faults, and lattice discreteness effects.
2. **Frozen Boundary Mantle ($\|\mathbf{X}_i - \mathbf{X}_{\mathbf{core}}\| > R_{\mathbf{free}}$)**: Nodes are held fixed under Dirichlet boundary conditions $\mathbf{x}_i = \mathbf{X}_i + \mathbf{u}_0(\mathbf{X}_i)$.

The Finite-Size Boundary Reaction Dilemma

While the analytical displacement field $\mathbf{u}_0$ satisfies traction-free conditions at infinity ($r \to \infty$), at any *finite* boundary distance $r = L/2$, the local stress field $\sigma_{ij}^{\text{analytical}} \propto \frac{\mu b}{2\pi(1-\nu)r}$ is non-zero. Freezing the mantle nodes prevents the finite body from undergoing the macroscopic strain necessary to accommodate the extra atomic half-plane.
Consequently, the crystal develops a macroscopic Cauchy stress tensor $\bar{\boldsymbol{\sigma}} \neq \mathbf{0}$, which distorts the core structure, shifts the apparent core energy, and creates spurious image forces that repel or attract the dislocation.

The objective of **Full Cauchy Tensor Macroscopic Stress Relaxation** is to determine the macroscopic deformation gradient $\mathbf{F}_{\text{ext}} \in \mathrm{GL}^+(2)$ such that the volume-averaged Cauchy stress tensor vanishes identically:

$$\bar{\boldsymbol{\sigma}}(\mathbf{F}_{\text{ext}}) = \frac{1}{\Omega} \int_{\Omega} \boldsymbol{\sigma}(\mathbf{x}) \, \mathrm{d}\Omega \longrightarrow \mathbf{0}. \tag{1}$$

2 Kinematic Parameterization: Why Upper-Triangular $\mathbf{F}_{\text{ext}}$?

A naive approach to macroscopic stress relaxation applies an unconstrained affine transformation or a symmetric stretch tensor $\mathbf{F} = \mathbf{U} = \mathbf{U}^T$ (with $F_{12} = F_{21}$). However, for an edge dislocation with horizontal Burgers vector $\mathbf{b} = (b, 0)^T$, symmetric shear imposes:

$$\Delta\mathbf{F}_{\text{sym}} = \begin{pmatrix} \Delta\varepsilon_{11} & \Delta\varepsilon_{12} \\ \Delta\varepsilon_{12} & \Delta\varepsilon_{22} \end{pmatrix}. \tag{2}$$

The off-diagonal term $(\Delta\mathbf{F})_{21} = \Delta\varepsilon_{12} \neq 0$ rotates a line originally parallel to the x-axis by an angle $\theta \approx \Delta\varepsilon_{12}$, creating an artificial rigid-body rotation of the slip plane:

$$y_{\text{new}} - y_{\text{core}} = \Delta\varepsilon_{12}(x - x_{\text{core}}) + (1 + \Delta\varepsilon_{22})(y - y_{\text{core}}). \tag{3}$$

For points on the glide plane ($y = y_{\text{core}}$), $y_{\text{new}} \neq y_{\text{core}}$ whenever $x \neq x_{\text{core}}$. The slip line is artificially tilted!

2.1 The Upper-Triangular Slip-Preserving Kinematics

To preserve the horizontal orientation of the slip plane ($y = \text{const}$) while permitting complete relaxation of normal strains ($\varepsilon_{xx}, \varepsilon_{yy}$) and shear strain ($\gamma_{xy}$), we restrict $\mathbf{F}_{\text{ext}}$ to the upper-triangular Borel subgroup:

$$\mathbf{F}_{\text{ext}} = \begin{pmatrix} \lambda_x & \gamma \\ 0 & \lambda_y \end{pmatrix}, \qquad (\mathbf{F}_{\text{ext}})_{21} \equiv 0. \tag{4}$$

This parameterization guarantees:

- **Slip Plane Invariance**: For any atom lying on the slip plane $Y = Y_{\text{core}}$, its deformed vertical coordinate is:
$$y - y_{\text{core}} = 0 \cdot (X - X_{\text{core}}) + \lambda_y (Y - Y_{\text{core}}) = 0. \tag{5}$$
The glide plane remains strictly horizontal and collinear with $\hat{\mathbf{e}}_1$.
- **Simple Shear Glide**: The off-diagonal term γ corresponds to simple shear along the Burgers vector direction $\mathbf{b} \parallel \hat{\mathbf{e}}_1$, exactly mirroring the physical mechanism of edge dislocation glide without spurious spin.

3 Macroscopic Cauchy Stress Tensor Formulation

In our nonlinear finite element framework, the crystal energy is evaluated element-by-element over triangular elements $e \in \mathcal{E}$.

3.1 Element Deformation and Metric Tensor

For each triangular element e with shape function derivatives $\nabla_{\mathbf{X}} \mathbf{N}^{(e)}$ in the reference crystal, the total deformation gradient $\mathbf{F}^{(e)}$ and right Cauchy-Green tensor $\mathbf{C}^{(e)}$ are:

$$\mathbf{F}^{(e)} = \mathbf{x}_e \cdot \nabla_{\mathbf{X}} \mathbf{N}^{(e)}, \tag{6}$$

$$\mathbf{C}^{(e)} = \mathbf{F}^{(e)T} \mathbf{F}^{(e)}. \tag{7}$$

To respect the discrete point-group symmetry $\mathcal{G} \subset \mathrm{GL}(2, \mathbb{Z})$ of the underlying crystal lattice (e.g., cubic/square or hexagonal/triangular), $\mathbf{C}^{(e)}$ is mapped into the fundamental Lagrange-reduced domain $\mathcal{D}_L$ via an integer unimodular matrix $\mathbf{m}^{(e)} \in \mathrm{GL}(2, \mathbb{Z})$:

$$\mathbf{C}_{\text{red}}^{(e)} = \mathbf{m}^{(e)T} \mathbf{C}^{(e)} \mathbf{m}^{(e)}. \tag{8}$$

3.2 First Piola-Kirchhoff and Cauchy Stress Tensors

The element strain energy density $\Phi(\mathbf{C}_{\text{red}})$ yields the second Piola-Kirchhoff stress in the reduced space:

$$\mathbf{S}_{\text{red}}^{(e)} = 2 \frac{\partial \Phi}{\partial \mathbf{C}_{\text{red}}^{(e)}}. \tag{9}$$

Pulling back via the reduction matrix $\mathbf{Z}^{(e)} = \mathbf{m}^{(e)}$ gives the first Piola-Kirchhoff stress $\mathbf{P}^{(e)}$:

$$\mathbf{P}^{(e)} = 2 \mathbf{F}^{(e)} \mathbf{Z}^{(e)} \frac{\partial \Phi}{\partial \mathbf{C}_{\text{red}}^{(e)}} \mathbf{Z}^{(e)T}. \tag{10}$$

The true spatial Cauchy stress $\boldsymbol{\sigma}^{(e)}$ is obtained by the standard push-forward:

$$\boldsymbol{\sigma}^{(e)} = \frac{1}{\det \mathbf{F}^{(e)}} \mathbf{P}^{(e)} \mathbf{F}^{(e)T}. \tag{11}$$

3.3 Volume-Averaged Macroscopic Stress

The crystal macroscopic Cauchy stress $\bar{\boldsymbol{\sigma}}$ is the area-weighted volume average over all elements:

$$\bar{\boldsymbol{\sigma}} = \frac{\sum_{e\in\mathcal{E}} \boldsymbol{\sigma}^{(e)}\Omega^{(e)}}{\sum_{e\in\mathcal{E}} \Omega^{(e)}}, \qquad \Omega^{(e)} = \det(\mathbf{F}^{(e)})\Omega_0^{(e)}. \tag{12}$$

Due to the symmetry of $\mathbf{C}$, $\bar{\boldsymbol{\sigma}}$ is strictly symmetric: $\bar{\sigma}_{xy} = \bar{\sigma}_{yx}$. In Voigt notation:

$$\bar{\boldsymbol{\sigma}}_{\text{voigt}} = \begin{pmatrix} \bar{\sigma}_{xx} \\ \bar{\sigma}_{yy} \\ \bar{\sigma}_{xy} \end{pmatrix} \in \mathbb{R}^3. \tag{13}$$

4 Instantaneous Spatial Tangent Stiffness ($\mathbb{C}^{\text{spatial}}$)

To achieve true quadratic Newton convergence when solving $\bar{\boldsymbol{\sigma}}(\mathbf{F}_{\text{ext}}) = \mathbf{0}$, we require the exact derivative of Cauchy stress with respect to macroscopic strain:

$$\mathrm{d}\bar{\boldsymbol{\sigma}} = \mathbb{C}^{\text{spatial}} : \mathrm{d}\boldsymbol{\varepsilon}. \tag{14}$$

In our code, this is evaluated directly by `AcousticTensor::getSpatialTangent()` in `src/acoustic_tensor.c`

4.1 Material Tangent Computation

At the current macroscopic state $\mathbf{F}_{\text{bar}} = \mathbf{F}_{\text{ext}}$, the acoustic tensor evaluates the fourth-order material tangent modulus $\mathbb{A}_{KLMN} = \frac{\partial^2\Phi}{\partial F_{kM}\partial F_{lN}}$ using the second derivatives $\frac{\partial^2\Phi}{\partial \mathbf{C}^2}$ calculated analytically by `BaseLatticeCalculator`.

4.2 Spatial Push-Forward and Geometric Stress Term

The push-forward to the spatial configuration consists of two essential contributions:

$$\mathbb{C}^{\text{spatial}}_{ijkl} = \underbrace{\frac{1}{J}F_{iK}F_{jL}F_{kM}F_{lN}\mathbb{A}_{KLMN}}_{\text{Term 1: Material Push-Forward}} + \underbrace{\frac{1}{2}\left(\sigma_{ik}\delta_{jl} + \sigma_{jk}\delta_{il} + \sigma_{il}\delta_{jk} + \sigma_{jl}\delta_{ik}\right)}_{\text{Term 2: Geometric Cauchy Stress Term}}, \tag{15}$$

where $J = \det(\mathbf{F}_{\text{bar}})$. The geometric stress term ensures rigorous consistency with the Jaumann rate of Cauchy stress under finite deformation.

4.3 Voigt Matrix Representation

The fourth-order spatial tensor $\mathbb{C}^{\text{spatial}}_{ijkl}$ is mapped into a 3×3 matrix $\mathbf{C}_{\text{voigt}}$:

$$\mathbf{C}_{\text{voigt}} = \begin{pmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1112} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2212} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1212} \end{pmatrix}. \tag{16}$$

5 The Newton-Raphson Inversion and Update Algorithm

At macroscopic iteration k, given the current residual stress $\bar{\boldsymbol{\sigma}}^{(k)}$ and tangent stiffness $\mathbf{C}^{(k)}_{\text{voigt}}$, we seek the strain increment:

$$\Delta\boldsymbol{\varepsilon} = \begin{pmatrix} \Delta\varepsilon_{11} \\ \Delta\varepsilon_{22} \\ 2\Delta\varepsilon_{12} \end{pmatrix}. \tag{17}$$

5.1 Solving the Linearized System

We solve:

$$\mathbf{C}_{\text{voigt}}^{(k)} \Delta\boldsymbol{\varepsilon} = -\bar{\boldsymbol{\sigma}}^{(k)}. \tag{18}$$

Using Eigen's full-pivoting LU decomposition (`fullPivLu`):

$$\Delta\boldsymbol{\varepsilon} = -\left(\mathbf{C}_{\text{voigt}}^{(k)}\right)^{-1} \bar{\boldsymbol{\sigma}}^{(k)}. \tag{19}$$

If $\det(\mathbf{C}_{\text{voigt}}) < 10^{-10}$ (near elastic instability or phase transition), a singular value decomposition (SVD) pseudoinverse is automatically invoked as a robust fallback.

5.2 Step Size Safeguard

To ensure unconditional stability even far from equilibrium, the strain increment is clamped:

$$\Delta\varepsilon_i \longleftarrow \operatorname{sign}(\Delta\varepsilon_i) \min\left(|\Delta\varepsilon_i|,\, \Delta\varepsilon_{\max}\right), \qquad \Delta\varepsilon_{\max} = 0.02. \tag{20}$$

5.3 Upper-Triangular Incremental Update

The incremental deformation gradient $\mathbf{F}_{\text{inc}}$ is constructed directly in upper-triangular form:

$$\mathbf{F}_{\text{inc}} = \begin{pmatrix} 1 + \Delta\varepsilon_{11} & 2\Delta\varepsilon_{12} \\ 0 & 1 + \Delta\varepsilon_{22} \end{pmatrix}. \tag{21}$$

The cumulative macroscopic deformation gradient updates multiplicatively:

$$\mathbf{F}_{\text{bar}}^{(k+1)} = \mathbf{F}_{\text{inc}}\, \mathbf{F}_{\text{bar}}^{(k)}. \tag{22}$$

Since both $\mathbf{F}_{\text{inc}}$ and $\mathbf{F}_{\text{bar}}^{(k)}$ have vanishing $(2,1)$ components, their product satisfies:

$$(\mathbf{F}_{\text{bar}}^{(k+1)})_{21} = 0 \cdot F_{11} + (1 + \Delta\varepsilon_{22}) \cdot 0 \equiv 0. \tag{23}$$

The upper-triangular structure is preserved *to machine precision at all steps.*

5.4 Core-Centered Atomic Coordinate Shift

All atomic coordinates $\mathbf{x}_j$ (both interior free nodes and boundary mantle nodes) are updated by an affine expansion *centered at the dislocation core* $\mathbf{x}_{\text{core}} = (x_c, y_c)^T$:

$$\mathbf{x}_j^{(k+1)} = \mathbf{x}_{\text{core}} + \mathbf{F}_{\text{inc}} \left(\mathbf{x}_j^{(k)} - \mathbf{x}_{\text{core}}\right). \tag{24}$$

Why Centering at $\mathbf{x}_{\text{core}}$ is Critical

If coordinates were scaled relative to the domain origin $(0,0)$, any non-zero shear $\Delta\gamma = 2\Delta\varepsilon_{12}$ or normal strain would translate the core position:

$$\mathbf{x}_{\text{core}} \longrightarrow \mathbf{F}_{\text{inc}}\mathbf{x}_{\text{core}} \neq \mathbf{x}_{\text{core}}.$$

This translation would misalign the core with the cylindrical relaxation window R_{free} and introduce artificial core drift. Centering the transformation at $\mathbf{x}_{\text{core}}$ guarantees that $\mathbf{x}_{\text{core}} - \mathbf{x}_{\text{core}} = \mathbf{0}$, anchoring the core invariant in space.

6 Nested Two-Scale Optimization Architecture

The full algorithm operates as a nested two-scale loop:

- **Inner Loop (Microscopic Relaxation)**: For a fixed outer mantle position (determined by the current $\mathbf{F}_{\text{bar}}$), L-BFGS minimizes the total energy with respect to the coordinates of the free interior atoms within R_{free}.
- **Outer Loop (Macroscopic Newton Step)**: Evaluates the volume-averaged Cauchy stress $\bar{\boldsymbol{\sigma}}$, computes $\mathbf{C}_{\text{voigt}}$, solves for $\mathbf{F}_{\text{inc}}$, and shifts the boundary positions.
- **Post-Convergence (Delaunay Remeshing)**: Once $\max|\bar{\sigma}_{ij}| <$ stress_tol, topological edge flips and Delaunay reconnections are tested around the core under zero macroscopic stress, accepting reconnections if and only if $\Delta E < 0$.

Algorithm 1 Full Cauchy Tensor Macroscopic Stress Relaxation

Require: Lattice $\{\mathbf{X}_i\}_{i=1}^{N}$, Core $\mathbf{x}_{\text{core}}$, Radius R_{free}, Tolerance η, Max Iters $K_{\max}$
Apply analytical Volterra / Stroh field: $\mathbf{x}_i \leftarrow \mathbf{X}_i + \mathbf{u}_0(\mathbf{X}_i - \mathbf{x}_{\text{core}})$
Partition nodes into free ($\leq R_{\text{free}}$) and Dirichlet boundary ($> R_{\text{free}}$)
Initialize $\mathbf{F}_{\text{bar}} \leftarrow \mathbf{I}$, converged $\leftarrow$ **false**
for $k = 0$ **to** $K_{\max} - 1$ **do**
    Inner Step: Relax free nodes via L-BFGS: $\mathbf{x}_{\text{free}} \leftarrow \arg\min E(\mathbf{x}_{\text{free}} \,|\, \mathbf{x}_{\text{bound}})$
    Compute volume-averaged Cauchy stress $\bar{\boldsymbol{\sigma}}$ via Eq. (12)
    $\sigma_{\max} \leftarrow \max(|\bar{\sigma}_{xx}|, |\bar{\sigma}_{yy}|, |\bar{\sigma}_{xy}|)$
    if $\sigma_{\max} < \eta$ **then**
        converged $\leftarrow$ **true**; **break**
    end if
    Compute $\mathbb{C}^{\text{spatial}}(\mathbf{F}_{\text{bar}})$ via `AcousticTensor::getSpatialTangent()`
    Extract 3×3 Voigt matrix $\mathbf{C}_{\text{voigt}}$ via Eq. (16)
    Solve $\mathbf{C}_{\text{voigt}} \Delta\boldsymbol{\varepsilon} = -\bar{\boldsymbol{\sigma}}_{\text{voigt}}$
    Clamp $\Delta\varepsilon_i \leftarrow \text{clamp}(\Delta\varepsilon_i, -0.02, 0.02)$
    Construct upper-triangular $\mathbf{F}_{\text{inc}}$ via Eq. (21)
    Update cumulative deformation: $\mathbf{F}_{\text{bar}} \leftarrow \mathbf{F}_{\text{inc}}\mathbf{F}_{\text{bar}}$
    Update all atom positions: $\mathbf{x}_j \leftarrow \mathbf{x}_{\text{core}} + \mathbf{F}_{\text{inc}}(\mathbf{x}_j - \mathbf{x}_{\text{core}})$
    Synchronize solver vectors and update element reference configurations
end for
if enable_remeshing **then**
    Run Delaunay topological reconnection loop on the stress-relaxed state ($\Delta E < 0$)
end if
Export relaxed state to VTK, CSV, and JSON

7 Numerical Verification and Finite-Size Scaling

7.1 Quadratic Convergence Demonstration

We illustrate the convergence behavior on a 25×25 square lattice with a central edge dislocation ($a = 1.0$, $R_{\text{free}} = 8.0$, Zanzotto continuum potential $\beta = -0.25, K = 5.0$):

- **Step 0 (Initial unrelaxed mantle)**:

$$E = 1.2041720450$$

$$\bar{\boldsymbol{\sigma}} = \begin{pmatrix} -0.051077 & -0.000020 \\ -0.000020 & -0.045825 \end{pmatrix}, \quad \max|\bar{\sigma}| = 5.11 \times 10^{-2}$$

$$\Delta\boldsymbol{\varepsilon} = (1.97 \times 10^{-3},\ 4.51 \times 10^{-4},\ 3.23 \times 10^{-6})^T$$

- **Step 1 (After 1 Newton step)**:

$$E = 1.1904803911 \quad (\Delta E = -0.01369)$$
$$\bar{\boldsymbol{\sigma}} = \begin{pmatrix} -4.68 \times 10^{-5} & -8.63 \times 10^{-6} \\ -8.63 \times 10^{-6} & -3.38 \times 10^{-5} \end{pmatrix}, \quad \max|\bar{\sigma}| = 4.69 \times 10^{-5}$$

Convergence reached in 2 iterations! The maximum stress dropped by over 1,000× in a single step, confirming second-order quadratic Newton convergence.

7.2 Finite-Size Scaling: Why 100×100 Behaves Differently

A key physical insight arises when comparing small (25×25) versus large (100×100) systems:

Grid Size	**Atoms**	**Unrelaxed** $\max|\bar{\sigma}|$	**Relaxed** $\max|\bar{\sigma}|$	**Box Expansion** $\Delta L/L$
25×25	625	5.11×10^{-2}	4.69×10^{-5}	+0.20%
100×100	10,000	4.40×10^{-4}	9.38×10^{-14}	+0.007%

Table 1: Comparison of boundary reaction stress and macroscopic relaxation across system sizes.

Because the initial analytical Stroh displacement field already represents an exact solution in an infinite domain, its boundary traction decays as $\sigma \sim \frac{1}{L}$. In a 100×100 crystal ($L = 100$), the initial residual stress is already very small (4.4×10^{-4}). The strain required to relieve it is merely $\sim 0.007\%$, which shifts the boundary by less than 0.007 lattice units. Consequently, the local core fitting parameters (E_{core}, K) are virtually unaffected by the distant boundary at $L = 100$, whereas for $L \leq 30$, macroscopic relaxation is vital to prevent severe artificial core compression.

8 Code Mapping and User Guide

The implementation spans three tightly integrated layers:

1. **C++ Physics Engine**:
   - `include/experiments/dislocation_study.h`: Added `relax_macro_stress`, `stress_tol`, `max_macro_iters`.
   - `src/experiments/dislocation_study.cpp`: Implemented the macro-stress Newton loop, spatial tangent extraction, upper-triangular update, and core-centered coordinate scaling.
   - `src/main.cpp`: Added CLI flags `--relax-stress=1`, `--stress-tol=1e-5`, and `--max-macro-iters=3`
2. **Data Logging**:
   - `dislocation_profiles.json`: Exports `macro_stress_relaxed`, `final_F`, and `final_macro_stress`.
3. **Lattice Launcher macOS GUI**:
   - `SimulationConfig.swift`: Added toggle, fields, and automatic `_relax` folder tagging.
   - `ContentView.swift`: Interactive UI toggle under *Mode & Triangulation*.
   - `DislocationProfileChartView.swift`: Dedicated KPI pills for **Macro Cauchy Stress** (showing $\max|\sigma|$ with green/red status) and **Macro Stretch F**.